DE–2511

Sub. Code
13

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2017.

DIGITAL COMPUTER FUNDAMENTALS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is gray code? Why it is important? Explain.

   (b) Subtract the following using 2's complement method

   (i) 1101 – 1001

   (ii) 110111– 11011.

2. (a) Explain the basic function of 3 to 8 line decoder with neat diagram.

   (b) Simplify the expression

   $S = ABCD + A'BCD + AB'CD + ABC'D + ABCD'$.

3. (a) How to add and subtract the numbers using parallel arithmetic element? Explain.

   (b) Explain the working concept of parallel Binary Adder.

4. (a) Discuss the purpose of Binary counters.

   (b) Explain the RS flip flop with its diagrammatic representation.

5. (a) What are the advanced concepts implemented in fifth generation computer? Explain.

   (b) Explain the basic concepts of computer hardware and software.

6. (a) Convert the following

   (i) $(1010)_2 = (\ )_{16}$

   (ii) $(0 \cdot 1011)_2 = (\ )_{16}$.

   (b) What is BCD code? What are its advantages and disadvantages?

7. (a) Explain the Master-Slave flip flops with neat diagram.

   (b) Discuss the basic concepts of clock.

8. (a) With diagram, explain shift right register.

   (b) Explain how computers are classified.

DE-2512

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2017.

DATA STRUCTURES USING C

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain briefly about data types in 'C' programming with suitable examples.

2. Discuss about passing arguments to function with suitable examples.

3. Explain briefly about dynamic memory allocation.

4. Discuss briefly about the stack structure and their operations with suitable examples.

5. What is linked list? Explain the concept of singly and doubly linked list.

6. (a) Discuss briefly about operators and expressions in C progamming.

   (b) Describe the following with suitable syntax

      (i) Switch

      (ii) Break

      (iii) Continue

      (iv) goto.

7. What is library function? Explain about passing a function.

8. What do you mean by arrays of pointers? Explain about passing functions to other functions.

DE–2513

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2017.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $(P \to Q) \wedge (R \to Q) \Leftrightarrow (P \vee R) \to Q$.

   (b) Prove that $\{\downarrow\}$ is functionally complete set.

2. (a) Show that $\neg Q \wedge (P \to Q) \Rightarrow \neg P$.

   (b) Prove that $\neg(P \uparrow Q) \Leftrightarrow \neg P \downarrow \neg Q$.

3. (a) Show that for any three sets A, B and C, $A \cap (B - C) = (A \cap B) - (A \cap C)$.

   (b) If $A = \{1,2,3\}$ and $B = \{a,b\}$ find $B \times A$ and A^2.

4. (a) Let R denotes a relation on the set of ordered pairs of positive $N \times N$ integers such that $\langle x,y\rangle R\langle u,v\rangle$ if and only if $xv = yu$. Show that R is an equivalence relations.

   (b) If R and S are relations from A to B, show that $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.

5. (a) Show that the inverse of an one-to-one and onto function is one-to-one and onto function.

(b) If $f : R \to R$ and $g : R \to R$ be any two function, where R is the set of real numbers, $f(x) = x^2 - 2$ and $g(x) = x + 4$. Find $g \circ f$ and $f \circ g$.

6. (a) If the binary operation defined as $a * b = \frac{ab}{2}$ for all $a, b \in R$, where R is the set of non zero real numbers. Find the inverse of a in R.

(b) Show that if every element in a group is its own inverse, then the group must be abelian.

7. (a) Find the identify element of the group of integers with the binary operation defined by $a * b = a + b + 1$ for all $a, b \in z$.

(b) Show that, A non empty subset S of a group G is a subgroup of $\langle G, * \rangle$ if and only if for any $a, b \in S$, $a * b^{-1} \in S$.

8. (a) How many vertices does a regular graph of degree 4 with 10 edges have?

(b) Show that the number of odd degree vertices in a group are even.

DE–9814

Sub. Code
16

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2017.

LAB I – (C AND DATA STRUCTURE)

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question is given to each candidate by lot system.

Each question has two subdivisions.

1. (a) Write a C program to input a temperature in Celsius and the find the corresponding temperature in Fahrenheit. (F = 9/5 *C + 32)

   (b) Write a C program to sort the given set of numbers using selection sort.

---------- Cut here ----------

2. (a) Write a C program to calculate Simple and Compound interest.

   (b) Write a C program to implement insert and queue operations on Queue using array concept.

---------- Cut here ----------

3. (a) Write a C program to convert a given number into words for 1 to 5. Ex. 1 to ONE, 2 to TWO and 5 to FIVE.

   (b) Write a C program to evaluate the given mathematical expression using stack.

4. (a) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS'. Otherwise 'FAIL'. Write a C program to do the above.

(b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

5. (a) Write a C program to input a basic pay and calculate Gross pay and Net pay.

INCOME:

(i) HRA = 15% of BASICPAY

(ii) DA = = 75% of BASICPAY

(iii) GROSSPAY = BASICPAY + DA + HRA

DEDUCTIONS:

(iv) PF = 10% of BASICPAY

(v) WCHARGE = 200

(vi) DEDUCTIONS = PF + WCHARGE

NETPAY = GROSSPAY + DEDUCTIONS

(b) Write a C program to implement insert and queue operations on Queue using array concept.

---------- Cut here ----------

6. (a) Write a C program to find the biggest of 5 numbers. Modify the program to find the biggest of 10 numbers.

(b) Write a C program to find an element using binary search.

7. (a) Write a C program to find square root, square and cube of any number.

   (b) Write a C program to sort the given set of numbers using Insertion sort.

---------- Cut here ----------

8. (a) Write a C program to find the biggest of 5 numbers. Modify the program to find the biggest of 10 numbers.

   (b) Write a C program to find an element using linear search.

DE-2514

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral)
DEGREE EXAMINATION, DECEMBER 2017.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the principles and benefits of object oriented programming. (10)

   (b) Explain the various data types supported by C++ with example. (10)

2. (a) Define the following : (10)

   (i) Object

   (ii) Class

   (iii) Data encapsulation

   (iv) Constructor

   (v) Destructor.

   (b) Explain the working of friend function with example. (10)

3. (a) Give the usage of new and delete operators in C++ with example. (10)

   (b) What is a this pointer? Explain with example. (10)

4. (a) Explain function over loading with example. (10)

   (b) Discuss about overloading unary operators with example. (10)

5. (a) Define Inheritance. Explain the various types of inheritance with example. (10)

   (b) What is a virtual function? Explain with example. (10)

6. (a) Discuss about compile time polymorphism and runtime polymorphism. (10)

   (b) Describe overloading Binary operators with example. (10)

7. (a) Discuss about inheritance access specifier. (10)

   (b) Explain dynamic constructor and copy constructor with example. (10)

8. (a) (i) Mention the pitfalls of operator overloading. (8)

   (ii) Define Reusability. (2)

   (b) Discuss the static and dynamic binding with examples. (10)

DE–2515

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Solve the following equations by Gaussian elimination method

$3x_1 + x_2 + x_3 = 4;$

$x_1 + 4x_2 - x_3 = -5$;

$x_1 + x_2 - 6x_3 = -12$

(b) Solve by Gauss seidal method $8x - 3y + 2z = 20;$ $6x + 3y + 12z = 35$ and $4x + 11y - z = 33$.

2. (a) Solve the following equations by Gauss-Jordan method $x + 2y + z = 8$, $2x + 3y + 4z = 20$; $4x + 3y + 2z = 16$.

(b) Estimate the value of $f(42)$ from the following data :

x :	20	25	30	35	40	45
$f(x)$:	354	332	291	260	231	204

3. (a) Use Lagrange's interpolation formula to find the value of y when $x = 10$.

x :	5	6	9	11
y :	12	13	14	16

(b) Evaluate $\int_0^5 \frac{dx}{4x+5}$ by Trapezoidal rule dividing the range into 10 equal parts.

4. (a) Find the first and second derivatives of $f(x)$ at $x = 1.5$

x :	1.5	2.0	2.5	3	3.5	4.0
y :	3.375	7.000	13.625	24.000	38.875	59.000

(b) Evaluate $\int_0^{\pi/2} \sin x \, dx$ by Simpson's rule dividing the range into 10 equal parts.

5. Obtain the mean, median and mode of the following :

Class :	10-25	25-40	40-55	55-70	70-85	85-100
f :	6	20	44	26	3	1

6. (a) Find the straight line that best fits the following data :

x :	1	2	3	4	5
y :	14	27	40	55	68

(b) The density function of a random variable X is given by $f(x) = Kx(2-x)$, $0 \le x \le 2$ find K, mean and variance.

7. (a) A random variable X has the probability function $f(x) = \frac{1}{2^x}$, $x = 1, 2, 3, \ldots$ find its M.G.F and hence find its mean.

(b) Memory capacity of a students was tested before and after a course of meditation for a month. State whether the course was effective or not from the data below :

Before :	10,	15,	9,	3,	7,	12,	16,	17,	4
After :	12,	17,	8,	5,	6,	11,	18,	20,	3

8. The following table gives monthly sales (in thousand rupees) of a certain firm in three states by its four salesmen.

	Salesmen			
States	I	II	III	IV
A	6	5	3	8
B	8	9	6	5
C	10	7	8	7

Setup the analysis of variance table and test whether is any significant difference (a) between sales by the firm salesmen (b) between sales in the three states.

DE–2516

Sub. Code
25

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) Lateral DEGREE EXAMINATION, DECEMBER 2017.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Briefly explain the following :

   (i) Windows desktop (5)

   (ii) Office shortcut bar. (5)

   (b) Write down the steps needed to create, open and save files in windows. (10)

2. (a) Explain the various commands for formatting text in MS-Word. (10)

   (b) Explain the different ways of page numbering in a word document. (10)

3. (a) How can you find and replace data in MS-Excel? (10)

   (b) How to format the data in MS-Excel? Explain. (10)

4. (a) Explain the text formatting and alignment in slide presentation. (10)

   (b) Discuss briefly about background and color scheme in MS-Power point. (10)

5. (a) Explain the procedure for creating and printing reports in MS-Access. (10)

   (b) Explain the procedure for creating and saving a table in MS-Access. (10)

6. (a) Explain the various printing commands in MS-Word. (10)

   (b) Write a note on spellcheck and mailmerge features in word. (10)

7. Write short notes on :

   (a) Moving from slide to slide in MS-Power point. (5)

   (b) Drawing on slides in MS-Power point. (5)

   (c) Adding speaker notes in MS-Power point. (10)

8. (a) How to enter formula in the formula bar in MS-Excel? Explain. (10)

   (b) What are the different types of charts in Excel? Explain. (10)

DE–9815

Sub. Code
26

DISTANCE EDUCATION

B.Sc. Computer Science/B.Sc. Computer Science (Lateral)
DEGREE EXAMINATION, DECEMBER 2017.

LAB II – (C++ PROGRAMMING LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.

Each sub division carries 50 marks.

1. (a) Write a menu driven C++ program to add and subtract the given two matrices of order m × n defined in class, using operator overloading.

   (b) Create a class called TIME that has integer data elements for hours, minutes, seconds. The constructors should initialize these data elements to specified value, if given, and otherwise to 0. A member function should display it, in 11:50:45 format. The final member function should add two objects of type Time passed as arguments.

2. (a) Assume you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number, and then generates the table, formatting it into ten columns and 20 lines.

(b) Write a program to process students marks with the help of classes. The class has private variables for name, mark1, mark2 and mark3. It has two member functions :

getdata( ) – to get input and result( ) – to print the results. All subjects mark must be > = 50 for pass otherwise Fail.

---------- Cut here ----------

3. (a) Using Pointers, create a class and write a program to get n names and display them in sorted order.

(b) Create a class Employee that contains employee number, employee name and address. Write a Menu driven C++ program to get the 'n' number of employee details and display all details in Employee name wise sorted order.

---------- Cut here ----------

4. (a) Write a program using Polymorphism to calculate the square of any two numbers of type int, float, double and long.

(b) Create a class DONOR that contains donor number, donor name, age, address, sex, and blood group. Write a Menu driven C++ program to display the number, name and address of the donors for the following categories :

(i) Blood donors having the blood group O+

(ii) Blood donors in the age group between 16 to 25

(iii) Female donors having blood group A in the age between 19 and 24.

5. (a) Write a program using friend function frifunc( ) which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

(b) Raising a number n to power p is the same as multiplying by itself p times. Write a function called power( ) that takes a double value for n and an int value for p, and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main( ) function that gets values from the user to test this function.

---------- Cut here ----------

6. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

(b) Create an equivalent for a four-function calculator. The program should request the user to enter a number, an operator, and another number. It should then carry out the specified arithmetical operations; adding, multiplying, subtracting, and dividing the two numbers. Finally it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

---------- Cut here ----------

7. (a) Write a C++ program to reverse the sentence and check whether find the given sentence is palindrome or not.

(b) Write a C++ program which will accept a string of 10 characters in length from the key board and count the occurrences of each of the five vowels in the string. The output p should be in a (tapped) format similar to this example.

A	E	I	O	U
0	1	0	0	1

8. (a) Using operator overloading, write a C++ program to find the different and total length of given two various tubes specified in meters and centimeters.

(b) Create a class Int to Overload all five integer arithmetic operators (+, ~, *, /, and %) so that they operate on objects of type Int. If the result of any such arithmetic operation exceeds the normal range of int's – from -32,768 to 32,767- have the operator print a warning and terminate the program. Write a program to test this class.

DE–9816

Sub. Code
27

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, DECEMBER 2017.

LAB III — APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question is given to each candidate by lot system.

Each question has two subdivisions.

1. (a) Prepare an invitation for a function to be conducted in your Institution. Use different text orientation and pictures to make it attractive in MS-WORD.

   (b) Create a STUDENT database with necessary tables and fields in MS-ACCESS and prepare a statement of MARKSHEET for students.

---------- Cut here ----------

2. (a) Prepare the First page of M.C.A PRACTICAL RECORD NOTE BOOK with picture insertion and alignment.

   (b) Create a Worksheet in MS-EXCEL for employee database with necessary fields.
   - (i) Include 5 records.
   - (ii) Calculate HRA = 20% of Basicpay
   - (iii) Calculate DA = 30% of Basicpay,
   - (iv) GROSSPAY = Basicpay+HRA+DA.
   - (v) Draw the bar chart between emp name and Grosspay.

3. (a) Prepare a news report using two columns, insert a picture in the first column and make the text flow around it in MS-WORD.

(b) Create Inventory table with fields like ITEMNO, NAME, QUANTITY ON HAND, REORDER LEVEL. Perform the following in MS-ACCESS.

(i) Create a form in custom format.

(ii) Create a query REORDER to show all the items.

(iii) Create a macro that executes REORDER query automatically.

---------- Cut here ----------

4. (a) Create two worksheets containing day to day house hold expenses for the months January and February 2015 with the following columns:

ITEM	AMOUNT
Stationery	25.00

(i) Add 10 different items as shown above, for two months in two different worksheets.

(ii) Consolidate both the months data and find the total expenses.

(iii) Find total expenses for two months.

(b) Prepare your resume in MS-WORlD with photograph inserted. Use table, bullets and different color features.

5. (a) Create a Worksheet with the following fields: Salesman number, Salesman Name, City, Product name, Sale Amount. Add three records for 5 different salesmen who have carried out sales of different products in different cities. Find the following :

   (i) Citywise Total Sales

   (ii) Salesmanwise total sales

   (iii) Productwise total sales

   (iv) Grand total sales.

   (b) Type lecture notes and provide audio explanation with the help of sound files in MS-WORD.

---------- Cut here ----------

6. (a) Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN and SALARY. Perform the following in MS-ACCESS

   (i) Add data for 20 employees.

   (ii) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI".

   (iii) Show all the records in the table for city "TRICHY" by filtering.

   (b) Create a form letter that informs the customer about the date of maturity of a deposit amount in commercial bank and request the customer for renewal. Mailmerge it with an Access. Prepare letters for customers whose due date falls in a specific range of dates.

7. (a) The following were the observations made in certain experiments for the values 'Y' and given the values of 'X'.

X:	1	2	3	4	5	6	7	8	9	10
Y:	10	20	45	25	16	36	27	15	30	40

Compute the following :

(i) Find the mean, median and standard deviation.

(ii) Correlation coefficient between X and Y.

(iii) Draw the bar chart and shading cells.

(b) Prepare your resume in MS-WORD with photograph inserted. Use table, bullets and different color features.

---------- Cut here ----------

8. (a) Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN and SALARY. Perform the following in MS-ACCESS

(i) Add data for 20 employees.

(ii) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI".

(iii) Show all the records in the table for city "TRICHY" by filtering.

(b) Create the Inventory worksheet in MS-EXCEL with necessary fields and perform the following:

(i) Enter all data items except QUANTITY ON HAND for 10 items.

(ii) Find total stock value in inventory as a product of total quantity on hand and total price. Display all items in red color whose quantity on hand is below recorder level.

DE–2517

Sub. Code
31

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Write different applications of computer graphics in detail. (10)

   (b) Write in detail about video display generator. (10)

2. (a) Explain DDA line drawing algorithm in detail. (10)

   (b) Write in detail about ellipse drawing algorithm. (10)

3. (a) Explain 2D transformation with matrix representation. (10)

   (b) Write briefly about 2D viewing transformations. (10)

4. (a) Explain convex polygon clipping in detail. (10)

   (b) Discuss any one line segment clipping in detail. (10)

5. (a) Describe 3D transformations in detail. (10)

   (b) Briefly discuss about projection plane and view volume. (10)

6. (a) Write about 3D clipping against a finite view volume in detail. (10)

   (b) Explain Z buffer algorithm in detail. (10)

7. (a) Discuss different styles of command language in detail. (10)

   (b) Explain the components of user interface. (10)

8. Write short notes on the following :

   (a) Point clipping (5)

   (b) 3D composite transformation (7)

   (c) Information display. (8)

DE–2518

Sub. Code
32

DISTANCE EDUCATION

B.Sc.(CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Differentiate between Character based system and Graphical User Interface. (10)

   (b) Describe the hardware and software required to get start with Windows. (10)

2. (a) Explain the concepts of Multitasking. (10)

   (b) Discuss in details about Multimedia Windows. (10)

3. (a) Explain the various operating modes on windows programming. (10)

   (b) Write notes on check box and Input box. (10)

4. (a) Explain the event-driven programming. (10)

   (b) Make a neat explanation about Paintbrush with examples. (10)

5. (a) Short notes on Drawing tools. (10)

   (b) Write short notes on multiple forms and startup forms. (10)

6. (a) Write short notes on OLE. (10)

   (b) Explain the IF and DO loops with suitable examples. (10)

7. (a) Write short notes on Windows power user. (10)

   (b) How to do coding and making EXE file. (10)

8. (a) Explain the Recursive function with suitable example. (10)

   (b) Elaborate the design of MDI forms. (10)

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DE–2519

Sub. Code
33

DISTANCE EDUCATION

B.Sc.(CS)/B.Sc.(CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

OPERATING SYSTEMS

(2007 onwards)

Time : 3 hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain in detail about structure of operating system.

2. Describe the following:

   (a) Round Robin Scheduling.

   (b) Message passing.

3. Explain I/O Devices and Device Controller with neat illustrations.

4. Discuss the following:

   (a) Deadlock prevention.

   (b) Deadlock avoidance.

5. Describe about various types of CPU scheduling algorithm.

6. (a) Discuss in detail about bitmap.

   (b) Explain the concept of Virtual memory.

7. (a) Discuss on file system reliability.

   (b) Explain the process of file system performance.

8. (a) Elucidate about Protection Mechanism with examples.

   (b) Describe the methods of Creating Page tables in detail.

DE–2520

Sub. Code
34

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Briefly discuss on TCP/IP.

   (b) Explain Client Server model in detail.

2. (a) Describe Domain Name System and Name Server Concepts.

   (b) Write about Internet E-mail.

3. (a) Explain in detail about Telnet (10)

   (b) Write short notes on :

      (i) URLs (5)

      (ii) Web Server. (5)

4. (a) Discuss the various applications of Electronic Commerce.

   (b) Explain the various resources available in Internet.

5. (a) Explain the features of object oriented programming concepts in JAVA.

   (b) Write a JAVA program to find the factorial of a given number.

6. (a) Explain the concepts of type casting with suitable examples.

   (b) Explain the basic data types used in Java with suitable examples.

7. (a) What is a constructor? Explain with an example.

   (b) What is inheritance? Explain different types of inheritance.

8. (a) Explain in detail about Applet Life Cycle.

   (b) Create an Applet to implement a bouncing ball.

DE–2521

Sub. Code
35

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS)(Lateral) DEGREE EXAMINATION, DECEMBER 2017.

RDBMS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) What are the advantages of DBMS over File Processing system? (10)

   (b) Discuss on manipulative capabilities of database schema. (10)

2. Explain the following with respect to Relational Model:-

   (a) Mapping operation. (10)

   (b) Data manipulation facilities. (10)

3. (a) Discuss on various types of databases using oracle. (10)

   (b) Discuss about relational database properties. (10)

4. (a) Describe the data access SQL commands and DML commands. (10)

   (b) Write queries for selecting distinct values and using sub queries in SQL. (10)

5. (a) How will you describe table definitions? Discuss with example. (10)

   (b) How is table modified and renamed? Explain with examples. (10)

6. (a) Write queries for using expressions and working with Null values in SQL. (10)

   (b) How will you join multiple tables in a query? Explain with examples. (10)

7. (a) How is table created? Explain with examples. (10)

   (b) Discuss on Embedded SQL. (10)

8. (a) Explain schema and subschema in detail. (10)

   (b) Describe the Data Control facilities in Relational Model. (10)

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DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, DECEMBER 2017.

LAB IV – VISUAL BASIC AND ORACLE

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.

Each question has two subdivisions.

1. (a) Write a VB program to read in principal, number of years and rate of interest through INPUT Boxes. Then, compute and print the simple interest and compound interest through labels.

(b) Create a table ADDRESS with the following structure.

Fieldname	Type	Width	Decimal
Name	Character	20	
Street	Character	20	
City	Character	20	
Pin	Numeric	6	

(i) Add 5 Records

(ii) Display the structure of the table

(iii) Add the field DISTRICT

(iv) Fill all DISTRICT with MADURAI

(v) Change the PIN to 630003 where city is KARAIKUDI and PIN is 623003.

2. (a) Write a VB program to compute the surface area and volume of a sphere for the given radius. Use Option buttons and INPUT Boxes.

(b) Create a table MARK with the following structure:

Fieldname	Type	Width	Decimal
REGNO	Numeric	8	
NAME	Character	20	
TAMIL	Numeric	3	
ENGLISH	Numeric	3	
MATHS	Numeric	3	
BIOLOGY	Numeric	3	

(i) Add 5 Records.

(ii) Show data in fields REGNO, NAME. TAMIL and BIOLOGY alone on the screen.

(iii) Show data in fields REGNO, BIOLOGY and MATHS alone on the screen.

(iv) Show data in fields NAME & REGNO alone on the screen.

(v) Show all fields on the screen.

3. (a) Write a VB program to compute the surface area and volume of a sphere for the given radius. Use Option buttons and INPUT Boxes.

(b) Create a table RESULT with the following structure.

Fieldname	Type	Width	Decimal
REGNO	Numeric	8	
NAME	Character	20	
MARK 1	Numeric	3	
MARK 2	Numeric	3	
MARK 3	Numeric	3	
Total	Numeric	3	
Result	Character	4	

(i) Add 5 records (Fill all Fields except Total and Result fields)

(ii) Fill TOTAL field with the sum of MARK 1, MARK 2 and MARK 3.

(iii) Fill the RESULT field with 'PASS' if TOTAL > = 150 otherwise 'FAIL'.

4. (a) Write a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using Label, Text and Option buttons.

(b) Create a table BIODATA with following structure,

Fieldname	Type	Width	Decimal
NAME	Character	25	
AGE	Numeric	2	
SEX	Character	1	
DEGREE	Character	8	
CITY	Character	20	

(i) Add 5 records, (BIODATA of 5 employees)

(ii) Display all records of MALE employees.

(iii) Delete all BSC degree holders and then display the table contents.

(iv) Remove all employee records whose AGE is greater than 20.

Display all records having AGE < 20 AND CITY is TRICHY.

5. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

(b) Create table LABEL with the following structure

Field name	Type	Width	Decimal
NAME	Character	20	
STREET	Character	20	
CITY	Character	20	
PIN	Numeric	6	

(i) Add 5 Records.

(ii) Display the contents of the table.

(iii) Display the Ascending order sorted list with NAME as primary key.

(iv) Display the Descending order sorted list with CITY as primary key.

(v) Display the contents with appropriate HEADINGS.

6. (a) Write a VB program to input two numbers and perform any one of the four arithmetic operations like addition, subtraction, multiplication and division as per user's choice.

(b) Create a table PERSONAL with the following structure.

Field name	Type	Width	Decimal
SNO	Numeric	8	
NAME	Character	20	
CITY	Character	20	
PIN	Numeric	6	
PHONE	Numeric	6	

(i) Add 5 Records

(ii) Display the contents of the table.

(iii) Display all names.

(iv) Display all names without duplicate.

(v) Display all names in uppercase, lowercase and Initial Capital letters.

(vi) Display all names and length of names.

(vii) Display name, city in which only beginning portion of name is known.

7. (a) Write a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using Label, Text and Option buttons.

(b) Create a table BLOOD with the following structure.

Field name	Type	Width	Decimal
DNO	Numeric	3	
DNAME	Character	20	
DOB	Date		
DAGE	Numeric	2	
DADD 1	Character	15	
DADD 2	Character	15	
DCITY	Character	15	
DPIN	Numeric	6	
DBLOOD	Character	3	
DSEX	Character	1	

(i) Add 5 Records.

(ii) Display all the records.

(iii) Display the information of donors in the age group 20 to 25 using BETWEEN option.

(iv) Display the information of donors having date of birth BETWEEN 01-JAN-55 and 31-DEC-75.

(v) Display the information of donors having any of the blood group from the set of blood groups using IN option.

(vi) Display the information of female donors with age between 20 and 25 using BETWEEN option.

8. (a) Write a VB program to compute and print either the SUM or the PRODUCT of the first N natural numbers. Use option button.

(b) Create a table NUMB with the following structure

Field name	Type	Width	Decimal
NUM	Numeric	6	2
ABSNO	Numeric	6	2
SQRTNO	Numeric	6	2
ROUNDNO	Numeric	6	2
TRUNCNO	Numeric	6	2
SQRNO	Numeric	6	2
CUBENO	Numeric	6	2

(i) Add 5 Records. (Some numbers must be negative or decimal)

(ii) Display all the records.

(iii) Replace ABSNO with absolute value of given numbers.

(iv) Replace SQRTNO with square root value of given numbers.

(v) Replace ROUNDNO by using ROUND function.

(vi) Replace TRUNCNO by using TRUNC function.

(vii) Replace SQRNO with square of given numbers.

(viii) Replace CUBENO with cube of given numbers.

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Sub. Code
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DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, DECEMBER 2017.

LAB V — JAVA PROGRAMMING LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question is given to each candidate by lot system.

Each question has two subdivisions.

1. (a) Write a Java program for sorting a given list of names in ascending order using command line arguments.

   (b) Write a Java program for handling Mouse events and Keyboard events.

---------- Cut here ----------

2. (a) Write a Java program to multiply two given matrices.

   (b) Write a Java program to count number of words and characters in a text.

---------- Cut here ----------

3. (a) Write Java programs for illustrating Overloading and Overriding Methods in Java.

   (b) Write a Java program to create and manipulate the following Text Areas, Canvas, Scroll bars, Frames, Menus and Dialog Boxes.

4. (a) Write Java programs for illustrating the implementation of various forms of Inheritance.(Ex. Single, Hierarchical, Multilevel inheritance ....)

   (b) Write a Java program to illustrate the use of files and Streams.

---------- Cut here ----------

5. (a) Write a Java program to illustrate the implementation of multilevel inheritance using interfaces in Java.

   (b) Write a Java program to display the number of characters, lines and words in a text file.

---------- Cut here ----------

6. (a) Write a Java program to illustrate the implementation of abstract class.

   (b) Write a Java program to display the contents of a file along with a line number at the beginning of each line.

---------- Cut here ----------

7. (a) Write a Java program to create Packages in Java.

   (b) Write a Java program to read and write the data using Random access file.

---------- Cut here ----------

8. (a) Write a Java program to Implement Producer/Consumer Problem using synchronization.

   (b) Write a Java program for creating Applets to draw the various polygons.